AdaSFMNet: Adaptive Spatial–Frequency Mamba for HSI–LiDAR Joint Classification

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***Abstract*—The joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) is an important task in remote sensing, as the complementary characteristics of the two modalities can significantly improve land cover recognition accuracy. However, existing methods mostly focus on spatial-domain feature modeling and lack sufficient exploration of frequency-domain information. Moreover, traditional wavelet transforms employ fixed filters, making it difficult to adaptively match the distinct frequency-domain characteristics of HSI and LiDAR. Meanwhile, these methods lack effective adaptation to frequency features, which limits fusion performance. To address these issues, this article proposes adaptive spatial-frequency mamba (AdaSFMNet), a joint classification network based on adaptive wavelet feature alignment and spatial–frequency Mamba (SF-Mamba). The proposed framework can adaptively extract spatial–frequency features while achieving global–local modeling across the spatial and frequency domains. For frequency-domain feature extraction, inspired by the discrete wavelet transform (DWT), learnable wavelet kernels are designed to adaptively separate the high-frequency and low-frequency components of HSI and LiDAR, and a cross-attention mechanism is employed to semantically align the high- and low-frequency components, thereby enabling fine-grained extraction and consistent representation tailored to the distinct frequency characteristics of each modality. To promote cross-domain fusion between the spatial and frequency domains, an SF-Mamba is introduced to fully exploit the spatial–frequency features of HSI and LiDAR. This design incorporates a spatial perception branch and a frequency filtering branch to complement the Mamba backbone with full spatial–frequency awareness, while also supporting both global and local modeling. Within AdaSFMNet, the adaptive wavelet decomposition and semantic alignment mechanism improve the extraction and alignment of multimodal features, while SF-Mamba removes redundant information through an adaptive spatial–frequency mapping branch and enhances the adaptation and fusion of cross-modal features in both the spatial and frequency dimensions. Their synergy produces more discriminative representations. Experimental results on three public datasets demonstrate the superiority of AdaSFMNet.**

***Index Terms*—Adaptive wavelet transform, hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR), spatial–frequency Mamba (SF-Mamba).**

I. Introduction

IN THE fields of land cover observation, urban development planning, and forest management, hyperspectral image (HSI) has gained widespread interest owing to its rich spectral information, which enables a relatively comprehensive acquisition of information from remote sensing images. However, HSI faces challenges in distinguishing objects made of the same material and is susceptible to external factors, such as cloud cover [1], [2]. To overcome these limitations, many other types of remote sensing data have been introduced to the HSI community [3], including multispectral imagery (MSI) [4] and light detection and ranging (LiDAR) [5]. Among them, LiDAR has received particular attention because it captures elevation information of land cover, which helps accurately to identify targets with small spectral differences and is insensitive to weather conditions. Given the complementary nature of the two modalities, LiDAR-derived elevation data provides critical object boundaries and terrain variations for spatial structure modeling, whereas HSI's fine spectral resolution facilitates the differentiation of materials with similar shapes but distinct compositions. Consequently, the joint use of HSI and LiDAR enables a multidimensional understanding of land cover, surpassing the performance limitations of single-source data. As a result, the joint classification of HSI and LiDAR has become a prominent research focus.

To better leverage the complementary cues in HSI and LiDAR data, researchers have developed a variety of fusion strategies [6], [7], [8], [9], [10]. These techniques generally fall into three broad paradigms: pixel-level fusion, feature-level fusion, and decision-level fusion. Specifically, pixel-level fusion methods are often associated with traditional machine learning approaches, using specific mathematical operations to merge raw images from different modalities [11]. Feature-level fusion models concentrate on integrating features extracted from raw images to form compact and discriminative representations [12]; their behavior depends on the feature extractor and fusion strategy. Decision-level fusion methods typically use preset or statistical weights to combine predictions from

Received 28 May 2026; revised 15 July 2026; accepted 21 July 2026. Date of publication 24 July 2026; date of current version 3 August 2026. This work was supported in part by Guangxi Key Technologies R&D Program under Grant AB2506910033, in part by Guangxi Science and Technology Major Program under Grant AA23073019, in part by the National Natural Science Foundation of China under Grant 62501177, in part by Guangxi Natural Science Foundation under Grant 2025JJB170042, in part by the Opening Fund of the Key Laboratory for Coastal Marine Eco-Environment Processes and Carbon Sinks of Hainan Province under Grant 2025SYKFKT05, and in part by the Project for Enhancing Young and Middle-Aged Teacher's Research Basis Ability in Colleges of Guangxi under Grant 2025KY0050. *(Corresponding author: Lianhui Liang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3716822

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different data sources to yield the unified classification decision. With the continuous development of deep models, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [13], the performance of feature-level fusion models has been significantly improved [14]. Considering the advantages of different fusion approaches, feature-level and decision-level fusion are often integrated to further boost both classification accuracy and model robustness. For example, in [15], a dual-branch CNN structure was proposed to capture features from HSI and LiDAR images, combining feature-level and decision-level fusion strategies during the fusion phase to integrate different types of features. As a milestone work in this field, coupled CNN [16] not only refreshed the accuracy records on several public datasets at the time but also laid a solid foundation for subsequent deep-learning-based fusion research. Building on this foundation, a large number of CNN-based fusion methods have emerged, further improving model performance by incorporating strategies such as adversarial complementary learning [16], multiscale feature fusion [17], [18], and cross-channel reconstruction [19]. Although the above CNN-based methods perform excellently in local feature extraction, the localized nature of convolutional operations limits their ability to model broad spatial contexts, hindering the capture of long-range dependencies in HSI spectral sequences and LiDAR spatial structures. In recent years, Transformer models based on the attention mechanism, with their global modeling capability, have achieved breakthroughs in natural language processing and computer vision and have gradually been introduced into multimodal remote sensing data fusion tasks. Yao et al. [20] addressed the multimodal adaptation of Vision Transformers by introducing ExViT, which extracts modality-specific features via separable convolutions and fuses them through cross-modality attention for land cover classification. Wang and Tan [21] reported a mutual benefit Transformer that achieves global mutual guidance in feature fusion, effectively boosting HSI–LiDAR fusion perception. To leverage the complementary strengths of CNNs and Transformers, Ding et al. [22] constructed a novel global–local Transformer network (GLT-Net), which combines a CNN for extracting multiscale local spatial features and a Transformer for learning global spectral features, thereby improving the classification performance of HSI and LiDAR data. Zhao et al. [23] designed a dual-branch framework for the fusion of hierarchical CNN and Transformer, effectively combining spectral–spatial representations from HSI with the elevation characteristics of LiDAR data, thereby boosting joint classification performance. Despite the considerable success of these Transformer-based approaches in global modeling, the intrinsic quadratic complexity of their self-attention mechanism incurs substantial computational burden when dealing with high-dimensional HSI and long sequence features, restricting their deployment in resource-limited settings [24]. In recent years, the Mamba architecture, built upon state-space models (SSMs), has surfaced in the vision community owing to its linear complexity and powerful capacity for modeling long-range dependencies [25]. Unlike Transformers, Mamba efficiently processes input sequences through a selective scan mechanism, significantly reducing the computational cost while maintaining the global perception. Motivated by this advantage, researchers have begun to introduce Mamba into HSI and LiDAR joint classification tasks. For instance, Gao et al. [26] constructed a multiscale feature fusion Mamba network that effectively extracts feature information from multimodal data by constructing spectral, spatial, and fusion Mamba modules, improving the performance while optimizing the computational efficiency. Wang et al. [27] presented a spatial–spectral SSM for HSI classification. This method processes spatial and spectral information through two parallel selective SSMs and employs a spatial–spectral mixing gate to dynamically fuse the extracted features, achieving competitive classification performance. Liao et al. [28] introduced HLMamba to address the challenge of fusing heterogeneous HSI and LiDAR features while exploiting the edge contour information. By combining a gradient joint algorithm with a multimodal Mamba fusion module, it efficiently captures long-range dependencies across modalities, improving the integration of spectral and elevation information. Nevertheless, the aforementioned methods are primarily confined to the spatial domain, with limited exploration of the frequency-domain features of ground objects, thereby restricting the full utilization of the representation potential of deep models.

Frequency-domain analysis, as a unique information perspective, can decompose an image into components of different frequencies, where high-frequency components correspond to edge and texture details of ground objects, and low-frequency components reflect their contour and structural information. Existing research has explored various frequency domain methods, which can be mainly divided into Fourier transform (FT)-based and wavelet transform-based approaches. Among them, the FT analyzes the entire spectrum of an image, focusing on global frequency-domain analysis and effectively extracting overall spectral features, but its ability to capture local frequency details is limited. Zhao et al. [29] introduced a fractional Fourier image transformer (FrIT) to address the tradeoff between global context modeling and local semantic preservation in HSI–LiDAR classification. By performing pixel-level fusion and learning contextual features across multiple fractional Fourier domains, FrIT efficiently captures both global and local representations. Zeng et al. [30] proposed a cross-modal hierarchical frequency-domain fusion method that adaptively fuses amplitude and phase information at different levels of different modalities, alleviating the issue of modality differences. Compared with the FT, the wavelet transform can provide both frequency- and spatial-domain details, thereby supplementing and enriching local details of images. Wang et al. [31] integrated the wavelet transform into the downsampling operation of neural networks and constructed a pure frequency-domain network, FDNet, which alleviates information loss caused by spatial downsampling. Ni et al. [32] further combined CNN, Transformer, and discrete wavelet transform (DWT), performing 2-D and 3-D DWT on HSI data and 2-D DWT on LiDAR data to extract high- and low-frequency features from each modality, then fused the multimodal features and designed a frequency-domain global feature encoder for classification. The FT excels at the global frequency-domain analysis

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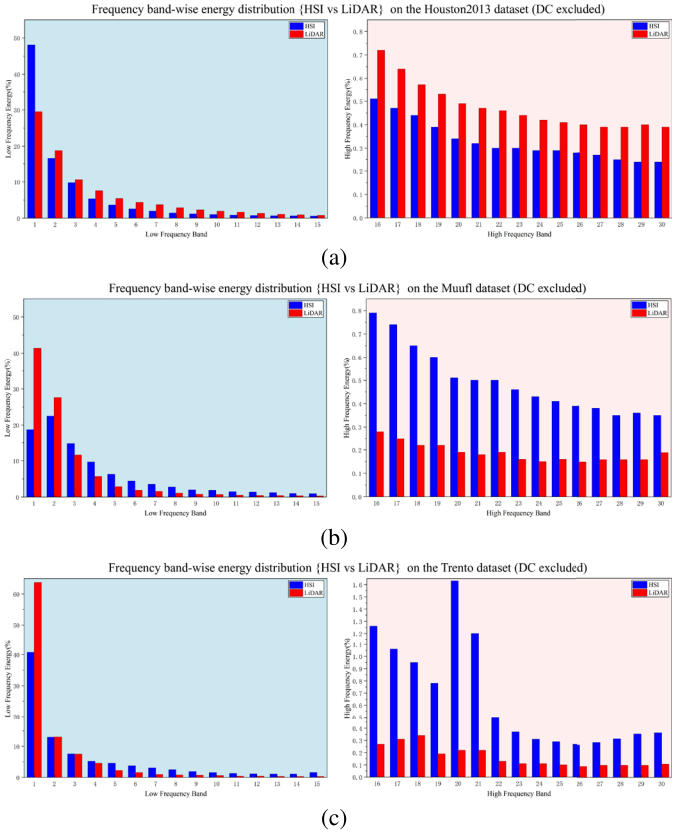


Fig. 1. Frequency bandwise energy distribution of HSI and LiDAR on three datasets in the Fourier domain (DC excluded). (a) Houston2013. (b) MUUFL. (c) Trento. The distributions vary across datasets: on Houston2013, HSI concentrates energy at low frequencies while LiDAR dominates mid-to-high frequencies; on MUUFL and Trento, the trend reverses. This demonstrates that the frequency spectra of HSI and LiDAR differ both within the same dataset and across different datasets, supporting the necessity of adaptive frequency-domain decoupling.

and is suitable for frequency filtering and global feature enhancement, while the wavelet transform, with its spatial–frequency localization capability, is more suitable for the feature extraction stage, providing a spatial-structure-rich frequency representation for subsequent fusion. In this article, the two work synergistically, each exploiting its own advantages.

Based on the above analysis, we investigate the challenges faced in HSI and LiDAR joint classification and propose the following improvement suggestions.

Exploration of Adaptive Frequency Decoupling: As shown in Fig. 1, the bandwise energy distributions of HSI and LiDAR exhibit scene-dependent variations across different datasets. On the Houston2013 urban dataset, HSI energy is predominantly concentrated in the lowest frequency bands, while LiDAR retains more energy in mid-to-high frequency bands, consistent with the abundant building edges and elevation discontinuities in urban scenes. However, on the MUUFL and Trento datasets, the global radial spectrum shows the opposite trend, with LiDAR exhibiting higher low-frequency energy ratios. This reversal can be attributed to the relatively flat terrain in these scenes, where LiDAR-derived elevation maps are naturally smooth and thus dominated by low-frequency components. Meanwhile, the HSI in these vegetation-rich scenes retains more mid-to-high frequency textures from tree canopies, grass patterns, and shadow boundaries, resulting in a higher proportion of high-frequency energy compared with LiDAR. This cross-dataset variability indicates that the frequency spectra characteristics of HSI and LiDAR are not fixed but depend on the specific scene structure [33]. Most existing frequency decoupling models, such as Fourier filters and wavelet transforms, rely on hand-crafted techniques that are not learnable, and their fixed filters cannot adaptively match the frequency-domain characteristics of complex remote sensing images. As a result, the obtained wavelet coefficients fail to fully encode the spatial structural information of ground objects, limiting the discriminability of subsequent features. In contrast to traditional fixed wavelet transforms, we parameterize the wavelet decomposition filters as learnable parameters and adopt modality-specific decomposition branches with non-shared weights for HSI and LiDAR. This enables the network to adaptively adjust its response to low- and high-frequency information according to the data distribution. The design inherently supports differentiated frequency band preferences for different modalities.

Exploration of spatial–frequency joint modeling and progressive fusion. After obtaining features rich in spatial–frequency information through adaptive frequency decoupling, how to further optimize these features to improve classification performance becomes crucial. Most existing methods directly perform fusion or classification after frequency decomposition, lacking further refinement of spatial–frequency mixed features, which leaves noise and redundant frequencies in the feature representation and limits further improvement of classification accuracy. Inspired by recent cross-domain fusion classification methods for HSI and LiDAR [34], [35], [36], we adopt a progressive fusion approach. Unlike single-step fusion, the multistage progressive fusion architecture maintains modality-specific features while alleviating conflicts between modalities, thereby generating a more stable and discriminative fused representation. Transformer-based models [37], [38] enhance feature interaction through stacking, while Mamba's SSM improves global information modeling through a selective mechanism and hardware-aware design, maintaining linear complexity. In fact, several Mamba-based methods for joint classification of HSI and LiDAR data have been developed [28], [39]. However, existing methods generally suffer from two key issues. On the one hand, they focus only on high-dimensional features in the spatial domain and lack effective adaptation to frequency features. Directly using these features often leads to loss of frequency information and degradation of fusion performance. On the other hand, Mamba lacks complete spatial and frequency perception, both of which are essential for the joint classification task. To address these two issues, we introduce a spatial–frequency Mamba (SF-Mamba) block, which integrates an adaptive spatial–frequency mapping module into the 2D-state-space duality (SSD) module of the Mamba2 architecture, further enhancing the adaptability and fusion capability of different frequency features.

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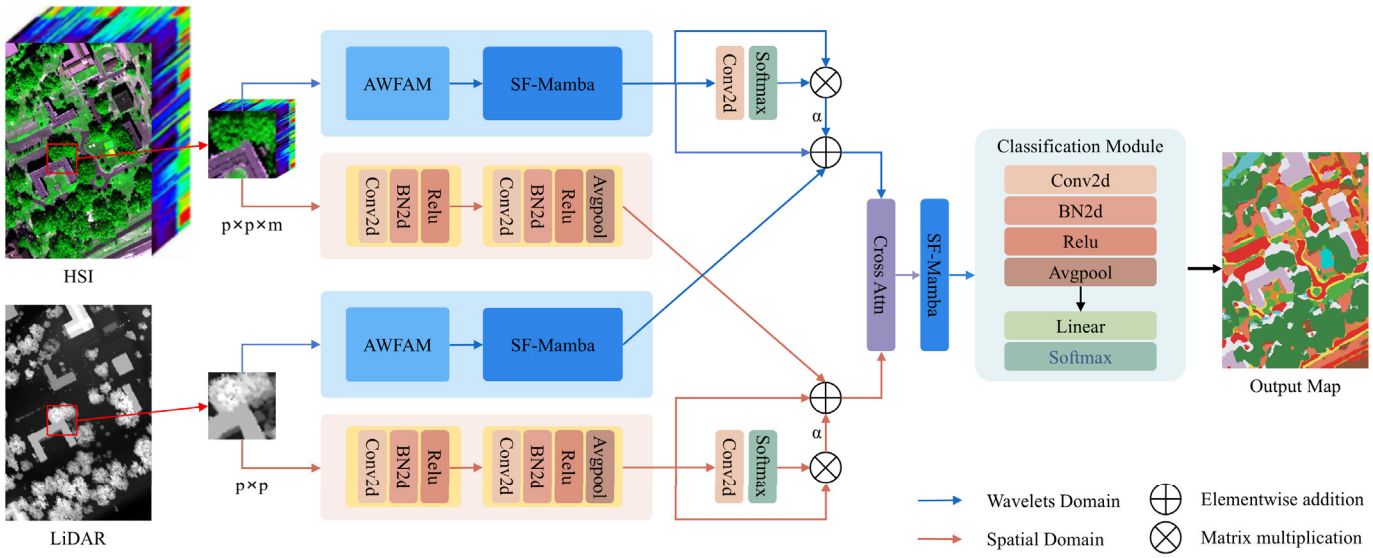


Fig. 2. Overall framework of the proposed AdaSFMNet for HSI–LiDAR joint classification. HSI and LiDAR images are first divided into patches of the same size and then processed by AWFAM for frequency-domain feature extraction and by a CNN for spatial feature extraction, respectively. The frequency-domain features are optimized by a single-layer SF-Mamba. Subsequently, cross-modal fusion is performed under the guidance of HSI features, followed by cross-attention fusion of spatial and frequency features, which are then refined by a two-layer stacked SF-Mamba. Finally, the fused features are fed into a classification module for joint classification.

Based on the above discussion, we propose the adaptive spatial-frequency mamba (AdaSFMNet) framework, as shown in Fig. 2, which consists of two key components: an adaptive approximate wavelet feature extraction module and an SF-Mambamodule. Specifically, for a pair of HSI and LiDAR images, AdaSFMNet first performs adaptive frequency decoupling, alignment, and fusion on HSI and LiDAR, respectively, through the adaptive wavelet feature alignment module (AWFAM), decomposing the images into multiband subsignals that represent low-frequency global information and high-frequency detail features, thereby forming a spatial–frequency joint representation that simultaneously encodes spatial location and frequency components. Since this type of mixed feature encoding both spatial position and frequency content is difficult for a generic Mamba model to handle directly, we introduce a specially designed SF-Mamba, which optimizes such features through parallel spatial perception and frequency filtering branches. Specifically, the spatial perception branch enhances local contextual dependencies, while the frequency filtering branch dynamically selects critical frequency components and suppresses noise, achieving clean global context modeling. The outputs of the two branches are fused to obtain an enhanced feature representation that combines local details and global structure with suppressed noise. This design enables SF-Mamba to provide cleaner, less redundant input for subsequent cross-modal fusion tasks. SF-Mamba dynamically adapts to different frequencies and modalities to achieve efficient fusion and can operate either on shallow features after frequency decoupling or on deep features after cross-domain fusion. On the other hand, we leverage the powerful spatial feature extraction capability of CNNs to capture spatial features from presegmented inputs. Then, to mitigate the adverse impact of excessive redundant information in HSI on fusion and classification, we adopt an HSI-dominant fusion mechanism with dynamically adjusted weights to realize adaptive fusion of HSI and LiDAR. For the multimodal fusion features in the frequency and spatial domains, a cross-attention mechanism is used to achieve global feature fusion.

In general, the primary contributions of this article can be summarized as follows.

1) A novel AdaSFMNet framework is proposed for the joint classification of HSI and LiDAR, which extracts joint spatial–frequency representations via AWFAM and progressively optimizes them using SF-Mamba, enabling end-to-end joint learning from feature extraction to cross-domain fusion.
2) A novel adaptive wavelet feature alignment approach is proposed for HSI and LiDAR joint classification. Inspired by the DWT, this approach parameterizes the wavelet decomposition filters as learnable parameters and adopts modality-specific decomposition with nonshared weights for HSI and LiDAR, achieving adaptive extraction of differentiated frequency-band features. Meanwhile, it semantically aligns low-frequency global information with high-frequency detail features, eliminating frequency-domain shifts across modalities and producing more consistent representations.
3) To the best of our knowledge, the SF-Mamba is introduced for the first time to the HSI–LiDAR joint classification task. This design facilitates efficient and synergistic cross-domain fusion through a spatial perception branch and a learnable frequency filtering branch. While preserving the linear complexity of the Mamba architecture, this dual-branch structure balances global and local receptive fields, enabling the extraction of both local details and long-range contextual information.

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4) Extensive experiments on three public multimodal remote sensing classification datasets demonstrate that AdaSFMNet outperforms existing state-of-the-art (SOTA) methods. Ablation studies further validate the individual effectiveness of the adaptive wavelet alignment approach and the SF-Mamba design as well as their synergistic contribution.

The remainder of this article is organized as follows. Section II reviews related literature on wavelet transform fusion methods and the development of SSMs. Section III describes the proposed AdaSFMNet model in detail. Section IV shows the comparison of AdaSFMNet with SOTA methods and conducts ablation studies to verify its effectiveness and analyze its performance. Finally, Section V summarizes the main contributions and research conclusion of this article.

II. Related Works

Significant progress in deep learning has greatly improved remote sensing image classification tasks. By leveraging robust feature extraction and generalization, deep learning surpasses traditional methods, such as regularized linear discriminant analysis [40], principal component analysis [41], and random forest [42]. Existing multimodal data fusion architectures mainly employ simple concatenation, elementwise addition, attention mechanisms, and Mamba models [43]. Therefore, this section focuses on two topics closely related to this article: adaptive frequency feature extraction and SSMs.

A. Adaptive Frequency Feature Extraction

Integrating frequency feature learning methods with architectures such as convolutional neural networks and Transformers has become a research hotspot in remote sensing image classification [32], [35]. Traditional frequency-domain decoupling methods, such as fast FT (FFT) and DWT, typically process frequency information in a static manner. Although these methods perform well in conventional classification tasks, their fixed frequency-domain decoupling schemes cannot dynamically adapt to the diversity and complexity of spectral distributions in multisource remote sensing images, limiting their applicability in complex scenarios. In contrast, recent studies have begun to explore adaptive frequency mechanisms within deep learning frameworks to achieve dynamic adjustment and decoupling of frequency information. For example, Zhou et al. [44] designed a position-related low-pass filter to adaptively extract low-frequency components from spatial features. Zheng et al. [45] proposed a phase-amplitude decoupling method that dynamically fuses high-frequency and low-frequency components using a frequency-adaptive multilayer perceptron (MLP). However, these methods are primarily oriented toward general remote sensing tasks and do not fully consider the essential differences in frequency band preferences between HSI and LiDAR. Therefore, how to design an adaptive frequency decoupling method tailored to the differentiated frequency band characteristics of multisource remote sensing data remains an open problem.

B. State-Space Models

In a continuous linear time-invariant system, the SSM establishes a mapping from the input $x(t) \in \mathbb{R}$ to the output $y(t) \in \mathbb{R}$ through a hidden state $h_t \in \mathbb{R}^N$. The evolution of the hidden state and the corresponding output can be expressed by a set of linear ordinary differential equations

$$\begin{aligned} \frac{dh(t)}{dt} &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix, $\mathbf{B} \in \mathbb{R}^{N\times 1}$ is the input matrix, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ is the output matrix.

To implement SSM in a deep learning framework, the continuous differential equations need to be discretized. Using the zero-order hold method, the system can be approximated as follows:

$$\begin{aligned} h_t &= \mathbf{A}h_{t-1} + \mathbf{B}x_t \\ y_t &= \mathbf{C}h_t \end{aligned} \tag{2}$$

where $\mathbf{A} = \exp(\Delta\mathbf{A})$ and $\mathbf{B} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$ are the discretized state parameters and Δ denotes the step size in the discretization process. For long sequences, efficient computation can be achieved by convolution with the kernel $\mathbf{K} = (\mathbf{CB}, \mathbf{CAB}, \ldots, \mathbf{CA}^{L-1}\mathbf{B})$, i.e., $y = x * \mathbf{K}$. This method leverages optimized convolution operations and reduces computational complexity. However, the model described by (2) is a linear time-invariant system whose parameters do not change with the input. To overcome the limitation of insufficient adaptability to varying inputs, Mamba [46] introduces a selective scan mechanism. This mechanism dynamically generates $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ from the input $x \in \mathbb{R}^{B\times L\times D}$, enhancing flexibility and efficiency in an input-dependent manner. On this basis, Mamba2 [47] introduces the SSD module, which parallelizes the mapping from $\{\mathbf{X}, \mathbf{A}, \mathbf{B}, \mathbf{C}\}$ to $\mathbf{Y}$ in a single projection, significantly improving the global modeling capability in a manner similar to the $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ projections in attention mechanisms. However, although Mamba performs well in multimodal tasks, its adaptability to the frequency characteristics of high-dimensional multimodal inputs and 2-D images remains limited. Existing SSD modules are primarily designed for 1-D time series signals, lacking complete spatial–frequency perception and thus struggling to effectively handle more complex spatial–frequency features.

III. Proposed Method

A. Overall Framework

The overall framework of the proposed AdaSFMNet is illustrated in Fig. 2. For the joint classification task of HSI and LiDAR data, given a pair of aligned source images, on one hand, AWFAM is used to perform frequency decoupling and alignment for each modality. The resulting features are then fed into the SF-Mamba module for shallow optimization and fusion of adaptive frequency-domain features. On the other hand, the powerful spatial feature extraction capability of CNNs is employed to capture spatial features from the input. To dynamically adjust the weights of HSI and LiDAR in cross-modal fusion and remove redundant information, we adopt an

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HSI-dominant fusion mechanism. Taking the spatial features $\mathbf{O_s^h}$ and $\mathbf{O_s^l}$ of HSI and LiDAR images as an example, the fusion process is expressed as follows:

$$\mathbf{O_s^f} = \mathbf{O_s^h} + \mathbf{O_s^l} - \alpha\left(\sigma\left(\mathrm{Conv}\left(\mathbf{O_s^h}\right) \times \mathbf{O_s^h}\right)\right) \tag{3}$$

where $\mathbf{O_s^f}$ denotes the fused spatial feature, $\sigma(\cdot)$ is the Sigmoid function, and α controls the strength of redundancy suppression. Intuitively, the sigmoid-gated convolutional term $\sigma(\mathrm{Conv}(\mathbf{O_s^h})) \times \mathbf{O_s^h}$ identifies spatially redundant information in HSI features through a soft mask mechanism: a larger α subtracts more HSI-dominant redundant information, allowing LiDAR features to play a more prominent role in the fused representation. Similarly, the cross-modal fusion feature in the wavelet domain, denoted by $\mathbf{O_w^f}$, is obtained by applying the same operation to the wavelet-domain features. Subsequently, a cross-attention mechanism is employed to fuse $\mathbf{O_s^f}$ and $\mathbf{O_w^f}$. Finally, SF-Mamba further refines the fused spatial–frequency mixed features, providing compact and discriminative collaborative representations for the final classification module. The two key components are described next: the AWFAM and the SF-Mamba module.

B. Adaptive Wavelet Feature Alignment Module

To address the problem that traditional wavelet transform has fixed spatial–frequency resolution in the task of HSI and LiDAR frequency decoupling and cannot adaptively match the frequency band differences between the two modalities, this article proposes an AWFAM. The framework of AWFAM is shown in Fig. 3. It performs domain transformation on multimodal image inputs and parameterizes the wavelet decomposition filters as learnable convolution operators, allowing the filter coefficients to be adaptively updated during training to achieve more effective frequency decoupling and feature extraction. In addition, it aligns the high- and low-frequency features into a unified semantic space to form a compact fused representation, thereby optimizing the subsequent joint classification of HSI and LiDAR.

The core idea of AWFAM is to mimic the construction of traditional wavelet transform and redefine signal decomposition based on learnable wavelet kernels, so as to automatically adapt to a wider range of signal processing needs [48]. In practice, the type of wavelet (e.g., Haar and Daubechies) is specified before training and determines the initial values of the learnable filters. To formally describe this construction, we first describe the wavelet transform for a 1-D signal. Given an input discrete signal t, scale factor j, time factor k, and scaling function ϕ_j, we define the wavelet function $\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$ and the scaling function $\phi_{j,k}(t) = 2^{j/2}\phi(2^j t - k)$. The decomposition of the signal at scale j_0 is

$$x(t) = \sum_{j>j_0}\sum_k w_{j,k}\psi_{j,k}(t) + \sum_k v_{j_0,k}\phi_{j_0,k}(t) \tag{4}$$

where $w_{j,k} = \langle x(t), \psi_{j,k}(t)\rangle$ are the detail coefficients, and $v_{j_0,k} = \langle x(t), \phi_{j_0,k}(t)\rangle$ are the approximation coefficients.

Next, we introduce adaptive analysis vectors $\vec{u}_0[k] = 2^{1/2}\phi(2t-k)$ and $\vec{u}_1[k] = 2^{1/2}\psi(2t-k)$ to replace the low-pass and high-pass filters. The decomposition of the signal in the wavelet basis can be viewed as a recursive convolution with specific filters

$$\begin{aligned} v_{j+1,p} &= \sum_k \vec{u}_0\left[k - 2p\right] a_{j,k} \\ w_{j+1,p} &= \sum_k \vec{u}_1\left[k - 2p\right] a_{j,k} \end{aligned} \tag{5}$$

where p is the sampling interval and $a_{j,k}$ is the value of the input signal at position (j,k).

For a 2-D input image feature $\mathbf{F_{m1}} \in \mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times m}$, the above decomposition is extended to

$$\begin{aligned} \mathbf{KF}_{LL} &= \vec{u}_0 \times \vec{u}_0^{\top}, \quad \mathbf{KF}_{LH} = \vec{u}_0 \times \vec{u}_1^{\top} \\ \mathbf{KF}_{HL} &= \vec{u}_1 \times \vec{u}_0^{\top}, \quad \mathbf{KF}_{HH} = \vec{u}_1 \times \vec{u}_1^{\top} \end{aligned} \tag{6}$$

where $\mathbf{KF}_{LL}$, $\mathbf{KF}_{LH}$, $\mathbf{KF}_{HL}$, and $\mathbf{KF}_{HH}$ represent the convolution operators for low-frequency, horizontal high-frequency, vertical high-frequency, and diagonal high-frequency, respectively, implemented via grouped convolution. The wavelet kernel is formed by concatenating these convolutional operators.

At this point, we have obtained the high- and low-frequency components of HSI and LiDAR. To obtain aligned features with enhanced representation consistency, we need to eliminate the inherent semantic differences between the high- and low-frequency components. First, we encode each component via depthwise convolution to obtain the low-frequency feature $\mathbf{X_L}$ and high-frequency feature $\mathbf{X_H}$. Then, we enhance the original features via weighted summation of multiscale convolutions. The enhanced low-frequency feature and the original high-frequency feature are fed into a cross-attention mechanism to promote semantic alignment between low- and high-frequency features. Specifically, we take the enhanced low-frequency feature $\widetilde{\mathbf{X_L}}$ as query and the original high-frequency feature $\mathbf{X_H}$ as key and value to compute the cross-attention output $\mathbf{Z_{L\to H}}$, allowing the low-frequency information to actively acquire detail and edge guidance from the high-frequency components. Symmetrically, we take the enhanced high-frequency feature $\widetilde{\mathbf{X_H}}$ as query and the original low-frequency feature $\mathbf{X_L}$ as key and value to obtain $\mathbf{Z_{H\to L}}$. This symmetric design enables the bidirectional exchange of complementary information between the two frequency branches: low-frequency features receive structural guidance from high-frequency details, while high-frequency features incorporate global contextual information from low-frequency contours. Through this mutual alignment, the frequency representations of HSI and LiDAR become better aligned, thereby alleviating the modality-specific frequency-domain shift and producing a more consistent joint representation. Finally, we perform the residual fusion of the two cross-attention outputs with their respective original features, generating an aligned feature $\mathbf{O_f}$ that preserves global structure while being rich in local details, providing a solid foundation for subsequent deep fusion and task decision-making.

C. Spatial Frequency Mamba Blocks

To overcome the limitations of Mamba, namely, its lack of adaptability to the frequency characteristics of high-dimensional, multimodal inputs and 2-D images, as well

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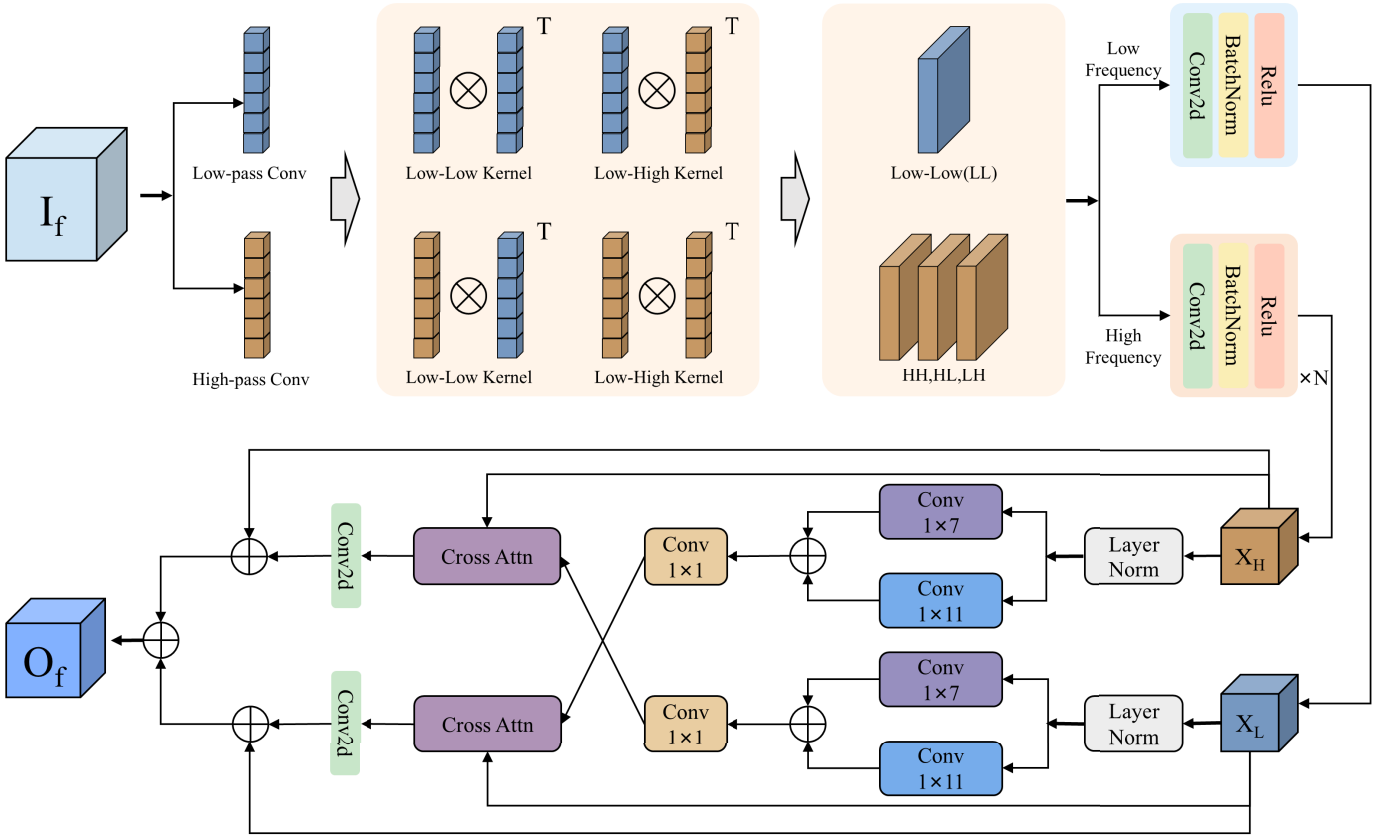


Fig. 3. Illustration of AWFAM. The module approximates wavelet decomposition using learnable low- and high-pass filters. The resulting high- and low-frequency components are first adjusted to the same channel dimension via convolution and then aligned to a unified spatial scale through two parallel convolutions with different kernel sizes. Cross attention is then applied to enable the interaction and selection between the two frequency features. Finally, the selected features are combined to produce the output.

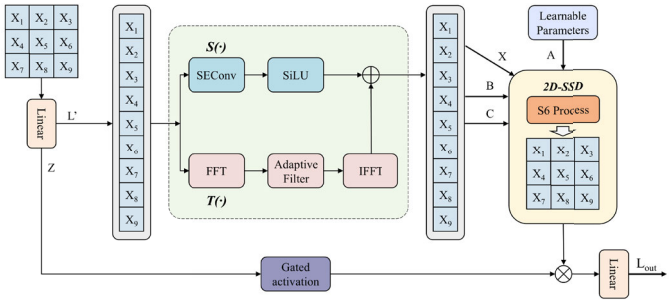


Fig. 4. Illustration of the SF-Mamba module. The input features are first linearly projected. Subsequently, the state modeling term is processed in parallel by a spatial perception branch and a frequency filtering branch. The outputs of the two branches are fused and fed into a 2D-SSD module for state-space modeling. Then, the result is combined with the gating term via elementwise multiplication, followed by a final linear projection.

as its insufficient spatial–frequency perception, we introduce the SF-Mamba block shown in Fig. 4. It relies on a key component—adaptive domain SSD (AdaD-SSD)—with two key improvements: 1) introducing spatial perception and frequency filtering modules and 2) updating the 2D-SSD architecture, thereby enhancing the ability to handle different frequency features.

1) Input and Projection: Specifically, given a normalized fused feature $\mathbf{L_{in}} \in \mathbb{R}^{H\times W\times m}$, AdaD-SSD first flattens it into a spatial sequence $\widetilde{\mathbf{L}} \in \mathbb{R}^{HW\times m}$. Then, through a linear projection layer, three parts are obtained (gating term $\mathbf{Z}$, state modeling term $\mathbf{L}'$, and step size term $\mathbf{\Delta_t}$)

$$\left[\mathbf{Z}, \mathbf{L}', \mathbf{\Delta_t}\right] = \widetilde{\mathbf{L}}\mathbf{W_{in}} \tag{7}$$

where $\mathbf{Z} \in \mathbb{R}^{HW\times m'}$, $\mathbf{L}' \in \mathbb{R}^{HW\times(m'+2Gd)}$, and $\mathbf{\Delta_t} \in \mathbb{R}^{HW\times N_h}$, $\mathbf{W_{in}}$ is the input linear projection matrix. m' is the expanded channel dimension, G is the number of groups, d is the state dimension, and N_h denotes the number of parallel state heads. The state modeling term $\mathbf{L}'$ is reshaped back to $\mathbf{L_{SF}} \in \mathbb{R}^{H\times W\times(m'+2Gd)}$ for spatial–frequency dual-branch enhancement.

2) Spatial–Frequency Dual-Branch Enhancement: To improve Mamba's spatial–frequency perception and global–local modeling capability, $\mathbf{L_{SF}}$ is simultaneously processed by a spatial perception branch $S(\cdot)$ and a frequency filtering branch $T(\cdot)$. For $S(\cdot)$, a 3×3 spatial convolutional operator W_{SE} combined with the sigmoid linear unit (SiLU) activation function $\sigma(\cdot)$ is used to enhance the spatial domain and dynamically adjust feature weights, thereby improving the local dependency and global representation of the fused features, yielding L_S

$$\mathbf{L_S} = \sigma\left(W_{SE}\left(\mathbf{L_{SF}}\right)\right) \in \mathbb{R}^{H\times W\times(m'+2Gd)}. \tag{8}$$

In $T(\cdot)$, $\mathbf{L_{SF}}$ is first transformed to the frequency domain Φ via FFT. To dynamically select frequency information

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beneficial for fusion, we adopt an adaptive filtering method that performs elementwise multiplication on the parameter set Λ and the 2-D frequency power spectrum. By learning an appropriate frequency threshold λ from the parameter set Λ, redundant information is effectively removed, and frequency components are optimized. Subsequently, the updated frequency-domain features are reconstructed back to $\mathbf{L_T}$ via inverse FFT (IFFT). The process can be modeled as follows:

$$\begin{aligned}\Lambda &= \mathrm{FFT}\left(\mathbf{L_{SF}}\right) \in \mathbb{R}^{H\times W\times(m'+2Gd)}\\ M &= \mathbf{1}\left(|\Lambda|^2 > \lambda\right) \in \{0,1\}^{H\times W\times(m'+2Gd)}\\ \mathbf{L_T} &= \mathrm{IFFT}\left(\Lambda \odot M\right) \in \mathbb{R}^{H\times W\times(m'+2Gd)}.\end{aligned} \tag{9}$$

Then, the spatial features $\mathbf{L_S}$ and frequency features $\mathbf{L_T}$ are fused via an adaptive cross-domain strategy. The fused representation is split into three components: state vector $\mathbf{X} \in \mathbb{R}^{HW\times m'}$, input matrix $\mathbf{B} \in \mathbb{R}^{HW\times Gd}$, and output matrix $\mathbf{C} \in \mathbb{R}^{HW\times Gd}$. In addition, $\mathbf{A}$ is input-independent and parameterized by learnable log-state coefficients, enabling state-space modeling

$$\mathbf{X},\mathbf{B},\mathbf{C} = \mathrm{Split}\left(\mathbf{L_S}\oplus\mathbf{L_T}\right), \quad \mathbf{Y} = \mathrm{SSD}\left(\mathbf{X},\mathbf{A},\mathbf{B},\mathbf{C}\right). \tag{10}$$

To better model the complex spatial structure of visual inputs, AdaD-SSD extends the original 1-D SSD framework to 2-D scenarios. By reshaping the flattened 1-D sequence into a 2-D feature map, the image is implicitly partitioned into multiple spatial subregions, each maintaining an independent local state. Through local neighborhood aggregation and cross-region global interaction, intraregion and interregion information exchange is achieved, endowing each token with complementary local and global spatial context. The SSM scan then proceeds along the inherent sequence order of the flattened feature map. The state update process follows a form similar to the 1-D version but incorporates spatial locality. The updated 2D-SSD equations can be rewritten from (4) as follows:

$$\begin{aligned}H_{ij,t} &= \mathbf{A_t}H_{ij,t-1} + \mathbf{B_t}X_{ij,t}\\ Y_{ij,t} &= \mathbf{C_t^T}H_{ij,t}\end{aligned} \tag{11}$$

where $H_{ij,t}$ denotes the hidden state at the 2-D coordinate (i, j) and time t, and $X_{ij,t}$ is the corresponding input feature at that position and time. $\mathbf{A_t}$ is the 2-D state transition matrix, and $\mathbf{B_t}$ is the 2-D input matrix, both acting on the hidden state and input features at the corresponding positions. During the update process, convolution operations in 2D-SSD capture fine-grained structures within subregions and global interactions between regions, thereby enabling the deep fusion of spatial and frequency information. This enhances detail expressiveness while improving structural consistency, resulting in the mapping from $\{\mathbf{X}_{2\text{-D}},\mathbf{A}_{2\text{-D}},\mathbf{B}_{2\text{-D}},\mathbf{C}_{2\text{-D}}\}$ to output $\mathbf{Y}_{2\text{-D}}$

$$\mathbf{Y}_{2\text{-D}} = \text{2D-SSD}\left(\mathbf{X}_{2\text{-D}},\mathbf{A}_{2\text{-D}},\mathbf{B}_{2\text{-D}},\mathbf{C}_{2\text{-D}}\right). \tag{12}$$

Finally, the output $\mathbf{L_{out}}$ of AdaD-SSD is obtained by linearly mapping the weighted result of the gated activation $\mathbf{Z}$ and $\mathbf{Y}$

$$\mathbf{L_{out}} = \left(\mathrm{LN}\left(\mathbf{Y}_{2\text{D}}\right)\odot\mathbf{Z}\right)\mathbf{W}_{out} \tag{13}$$

where $\mathbf{W_{out}}$ is the output linear projection matrix. Notably, we retain an MLP layer [49] similar to the Transformer block for feature updates.

D. Classification and Loss Function

The final classification module consists of two stages. First, a 1×1 convolution is applied to the fused features for channel compression, followed by batch normalization, rectified linear unit (ReLU) activation, and adaptive average pooling to obtain a compact global semantic vector. Second, this vector is fed into a fully connected layer for class mapping, and after softmax normalization, the probability distribution of the sample over each class is output. To alleviate the class imbalance problem, the classification loss adopts a weighted focal loss

$$\mathcal{L}_{\text{foc}}\left(P,Q\right) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{CLS}\alpha_i\, Q_{n,i}\left(1-p_{n,i}\right)^{\gamma}\log\left(p_{n,i}\right) \tag{14}$$

where N is the batch size, CLS is the number of classes, $p_{n,i}$ is the predicted probability that the nth sample belongs to class i, $Q_{n,i} \in \{0, 1\}$ is the one-hot ground-truth label, $\gamma \geq 0$ is the focusing parameter, and α_i is the weight for class i.

IV. Experiments and Analysis

In this section, we provide a comprehensive overview of the datasets, evaluation metrics, and other experimental settings used in our work. In addition, we conduct comparative experiments and ablation studies to demonstrate the effectiveness and superiority of AdaSFMNet.

A. Dataset Description

1) *Houston2013 Dataset:* As a standard benchmark for urban remote sensing, the Houston2013 dataset provides paired HSI–LiDAR data with precise geospatial alignment. The hyperspectral cube captures 144 spectral channels, while the LiDAR-derived digital surface model (DSM) provides complementary vertical structure information, both with dimensions of 349×1905 pixels [50]. There are 15 029 validated reference samples spanning 15 urban land cover categories.
2) *MUUFL Dataset*: The MUUFL Gulfport dataset (2010, MS, USA) combines HSI from the ITRES CASI-1500 sensor (64 bands, 380–1050 nm) and LiDAR from the ALTM system (1064 nm). The HSI has anisotropic resolution (0.54 m cross-track $\times$ 1.0 m along-track), while the LiDAR product has a resolution of 0.60 $\times$ 0.78 m. Both share spatial dimensions of 325 $\times$ 220 pixels and include 53 687 georeferenced samples over 11 land cover categories. Classification statistics are presented in [51].
3) *Trento Dataset*: The 2015 Trento dataset provides co-registered HSI and LiDAR data for precision agriculture. The HSI cube (AISA Eagle sensor) includes 63 bands (420–990 nm), and the LiDAR-derived DSM (Optech ALTM 3100EA) provides structural data at 1-m resolution. It contains 30 214 validated samples over six classes and is a key benchmark for agricultural remote sensing [52].

B. Experimental Setting

Our experiments were conducted on a server equipped with four NVIDIA Tesla V100-SXM2 (16 GB) GPUs and an

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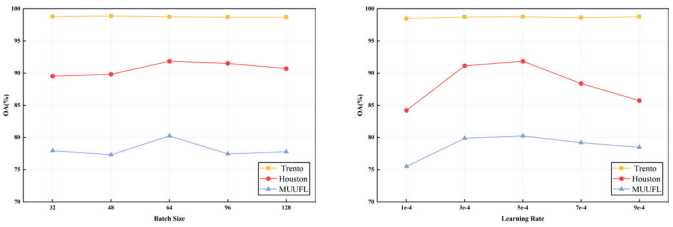


Fig. 5. Parameter sensitivity analysis on the Trento, Houston, and MUUFL datasets. (a) Batch size. (b) Learning rate.

Intel Xeon CPU E5-2696 v4. The implementation was carried out in PyTorch, using the Adam optimizer with a learning rate decay of 0.9. Based on empirical validation, we set the batch size to 64, the learning rate to 0.0005, and trained for 500 epochs, with evaluation every 20 epochs. Preprocessing involved generating patches of size $p \times p$ pixels. For all three datasets, ten training samples per class were randomly selected following the standard protocol. Model performance was evaluated using four metrics: average accuracy (AA), per-class accuracy (CA), Kappa coefficient, and overall accuracy (OA). All metrics were averaged over five independent trials with different random seeds to ensure robustness and fairness.

C. Parameter Analysis

To evaluate the robustness of AdaSFMNet with respect to key training hyperparameters, we conducted a sensitivity analysis on all three datasets by varying the batch size and learning rate while keeping other parameters fixed.

1) Analysis of the Batch Size: We evaluated candidate values {32, 48, 64, 96, 128}. As shown in Fig. 5(a), the OA on the Houston and MUUFL datasets increases as the batch size grows from 32 to 64, indicating that a moderate batch size leads to more stable gradient estimation and improved feature learning. When the batch size further increases to 96 and 128, the performance exhibits a downward trend. This suggests that overly large batch sizes reduce the stochasticity of optimization and weaken generalization. The Trento dataset shows relatively stable performance across all tested batch sizes, with OA varying within only 0.17%, further confirming the robustness of AdaSFMNet. Overall, a batch size of 64 provides a good balance between training stability and classification accuracy and is adopted in all our experiments.

2) Analysis of the Learning Rate: We evaluated candidate values $\{1e^{-4}, 3e^{-4}, 5e^{-4}, 7e^{-4}, 9e^{-4}\}$. As shown in Fig. 5(b), the OA on all three datasets improves as the learning rate increases from $1e^{-4}$ to $5e^{-4}$, indicating that a moderately larger learning rate facilitates more effective convergence. When the learning rate further increases beyond $5e^{-4}$, the performance begins to decline, with a more pronounced drop on the Houston and MUUFL datasets. This indicates that excessively high learning rates can reduce the stability of the optimization process. The Trento dataset remains relatively stable even at $9e^{-4}$, which is consistent with its simpler scene structure and higher class separability. Overall, a learning rate of $5e^{-4}$ achieves the best or near-best performance across the datasets and is adopted in all our experiments.

D. Performance Comparison

In this section, we describe the comparison of different classification methods across the three datasets.

1) Comparison of Different Classification Methods: To comprehensively evaluate the proposed AdaSFMNet, we selected eight SOTA multimodal remote sensing classification methods for comparison. These methods include GLT-Net [22], DSHFNet [53], M2FNet [54], FDNet [32], MICF-Net [55], S3F2Net [56], IDNet [34], and CHNet [57]. Specifically, we hold the following.

1) *GLT-Net:* Designed for joint HSI and LiDAR classification, this global–local Transformer network uses a CNN to extract local spatial features, a Transformer to model global spectral dependencies, and probabilistic decision-making to fuse multiscale features.
2) *DSHFNet:* To overcome the unreliability of single-scale features and the high dimensionality of multiscale representations, a dynamic scale hierarchical fusion network is proposed. It computes scale-space similarity, dynamically selects appropriate scales and applies spatial and modal attention modules for hierarchical fusion.
3) *M2FNet:* The number of parameters is significantly reduced by replacing standard self-attention with a dilated convolution-based encoder (DConvformer). The network also employs multiscale 3-D–2-D hybrid CNNs and a lightweight attention-free fusion module.
4) *FDNet:* Operating in the frequency domain, this method introduces an FFTtransform-based self-attention mechanism and convolutional offsets. It captures both local frequency features and global semantic features while lowering computational complexity.
5) *MICF-Net:* High-level semantic interaction and dynamic balance of multichannel data are achieved through dual-branch cross-channel attention transformers, adaptive mask modulation, and semantic-guided feature fusion.
6) *S3F2Net:* Combining CNN with graph convolutional networks, this approach constructs a graph from LiDAR data to simulate topological relationships of ground objects. A dynamic node update strategy integrates data representation from global and local perspectives.
7) *IDNet:* Laplacian high-frequency features are embedded into a Transformer. Dynamic distance-aware and similarity-based grouping interaction modules are designed, and differential convolution constructs detail enhancement experts to strengthen high-frequency detail discrimination.
8) *CHNet:* Using a coarse-to-fine high-order network, redundancy removal and feature reconstruction are performed in both frequency and spatial domains. Gating mechanisms, deep expert systems, and high-order feature statistics enable selective fusion and discriminability enhancement.

Tables I–III summarize the sample classification performance across the three datasets. The tables adopt a standardized format, with the best result in each class highlighted in bold, facilitating direct comparison of method effectiveness under a consistent evaluation protocol. Figs. 6–8 present

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TABLE I

QUANTITATIVE RESULTS ON HOUSTON2013 DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS ARE IN BOLD

Class	Classification accuracy achieved by different models (in %)								
	GLT-Net	DSHFNet	M2FNet	FDNet	MICFNet	S3F2Net	IDNet	CHNet	Proposed
C1	92.09	90.91	94.94	**97.90**	96.12	86.56	91.88	92.60	93.97
C2	90.00	86.86	82.23	86.64	85.77	76.46	85.82	**91.06**	83.38
C3	97.35	99.10	99.01	96.74	99.74	99.18	98.25	**99.77**	98.72
C4	93.87	**96.09**	92.84	93.47	94.17	89.82	92.97	92.90	95.72
C5	98.52	97.22	95.70	97.16	98.10	99.06	**99.74**	99.12	99.72
C6	83.81	85.78	81.46	85.97	**87.30**	86.67	86.48	83.62	83.87
C7	94.59	75.58	93.43	83.08	**97.41**	92.26	80.33	82.27	96.04
C8	77.10	74.31	81.52	78.18	85.11	**86.26**	85.80	83.34	85.33
C9	83.25	51.63	83.00	79.81	**91.47**	87.25	74.22	74.81	88.66
C10	92.97	85.26	83.01	76.35	83.43	39.85	89.42	81.77	**95.83**
C11	88.98	81.40	86.84	73.67	90.84	**94.04**	87.23	88.83	89.53
C12	94.24	78.05	85.69	69.47	75.72	41.80	**95.57**	94.85	83.35
C13	94.60	88.58	91.63	90.98	82.57	**98.65**	98.21	88.10	95.60
C14	97.85	94.35	93.92	96.99	**99.90**	97.70	99.38	99.00	99.81
C15	99.78	**99.97**	97.48	99.48	99.69	98.40	**99.97**	99.85	93.82
OA	91.46±1.07	83.96±3.19	89.01±0.66	85.56±1.50	90.74±0.92	82.44±3.47	89.82±0.57	89.40±1.26	**91.85±0.71**
AA	91.93±0.88	85.67±2.50	89.51±0.42	87.06±1.20	91.16±0.90	84.93±2.74	91.02±0.53	90.13±1.05	**92.22±0.55**
Kappa	90.77±1.16	82.66±3.45	88.12±0.71	84.39±1.62	89.98±0.99	81.02±3.74	88.99±0.62	88.53±1.36	**91.19±0.77**

TABLE II

QUANTITATIVE RESULTS ON MUUFL DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS ARE IN BOLD

Class	Classification accuracy achieved by different models (in %)								
	GLT-Net	DSHFNet	M2FNet	FDNet	MICFNet	S3F2Net	IDNet	CHNet	Proposed
C1	84.88	76.36	81.63	84.81	85.42	**90.06**	75.63	78.67	86.77
C2	67.25	65.97	59.65	71.54	59.11	66.84	61.82	64.62	**71.87**
C3	69.39	47.80	50.41	53.50	56.45	**69.06**	63.87	64.64	65.30
C4	56.95	62.02	40.72	53.38	55.06	52.22	66.93	**79.85**	59.85
C5	**81.13**	73.44	72.48	78.92	69.19	72.35	78.69	78.00	79.16
C6	99.96	75.39	99.39	99.39	99.91	96.27	98.16	96.54	**100.00**
C7	83.53	**88.26**	71.17	83.48	80.43	76.84	80.84	83.34	83.88
C8	89.34	83.84	84.82	86.12	81.16	61.52	86.85	59.72	**91.05**
C9	55.99	30.12	48.92	**71.42**	60.70	42.62	45.85	63.90	41.60
C10	49.02	12.95	50.52	53.76	**69.94**	49.71	49.83	65.55	56.18
C11	96.06	97.61	96.14	94.67	95.60	95.37	96.45	**99.61**	96.14
OA	79.87±0.42	71.09±12.54	72.57±1.06	77.76±3.10	75.35±1.54	76.89±0.39	74.08±2.14	73.53±1.12	**80.25±1.31**
AA	75.77±1.73	64.89±16.88	68.71±1.80	75.54±2.56	73.91±0.94	70.26±1.69	73.17±1.84	**75.86±0.63**	75.62±2.02
Kappa	74.24±0.48	63.48±16.23	65.31±1.18	71.71±3.71	68.58±1.83	69.77±0.20	67.58±2.44	66.63±1.30	**74.62±1.40**

TABLE III

QUANTITATIVE RESULTS ON TRENTO DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS ARE IN BOLD

Class	Classification accuracy achieved by different models (in %)								
	GLT-Net	DSHFNet	M2FNet	FDNet	MICFNet	S3F2Net	IDNet	CHNet	Proposed
C1	87.75	**99.93**	96.13	99.44	94.84	99.68	99.40	97.21	98.21
C2	95.06	**97.45**	79.94	87.17	95.93	93.46	93.82	94.36	90.67
C3	84.39	90.83	91.73	95.01	**97.87**	93.13	95.10	97.01	95.35
C4	**100.00**	99.79	98.57	99.86	99.92	99.75	99.98	98.94	**100.00**
C5	**100.00**	99.43	99.05	97.24	97.09	99.95	98.41	99.90	**100.00**
C6	96.02	90.39	93.45	92.52	97.06	91.30	95.21	96.74	**99.56**
OA	97.23±4.48	98.33±0.67	95.98±0.94	96.83±0.86	97.54±1.54	98.22±0.24	98.19±0.39	98.34±0.51	**98.75±0.28**
AA	93.87±9.51	96.30±1.95	93.14±1.88	95.21±0.50	97.11±0.93	96.21±0.50	96.99±0.58	**97.36±0.48**	97.30±0.62
Kappa	96.23±6.15	97.77±0.90	94.64±1.26	95.78±1.13	96.73±2.03	97.62±0.33	97.59±0.51	97.79±0.68	**98.33±0.37**

the standardized visualization protocol for all datasets, showing in order: (a) hyperspectral false-color images, (b) LiDAR visualizations, (c) ground truth maps, and (d)–(l) comparative classification visualizations. This systematic presentation

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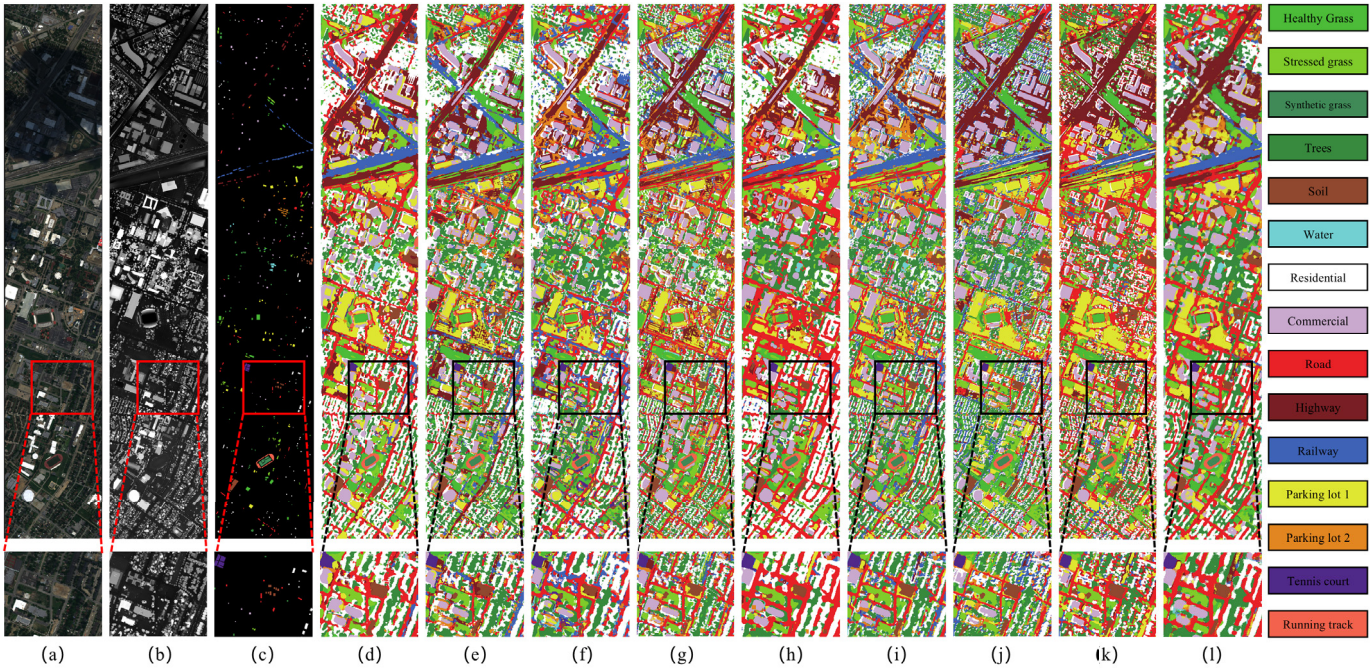


Fig. 6. Classification maps for all pixels in the Houston2013 dataset using different methods. (a) False-color HSI (using bands 64, 43, and 22, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) GLT-Net (91.46%). (e) DSHFNet (83.96%). (f) M2FNet (89.01%). (g) FDNet (85.56%). (h) MICF-Net (90.74%). (i) S3F2Net (82.44%). (j) IDNet (89.82%). (k) CHNet (89.40%). (l) AdaSFMNet (91.85%).

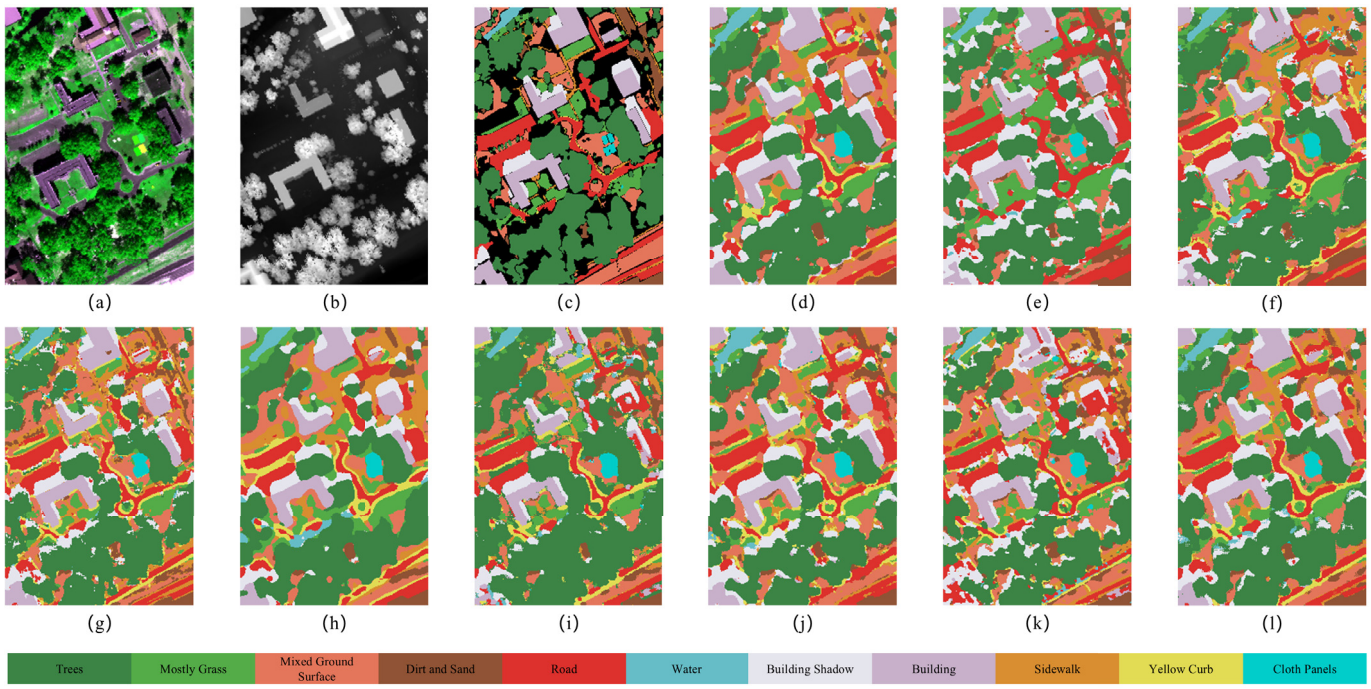


Fig. 7. Classification maps for all pixels in the MUUFL dataset using different methods. (a) False-color HSI (using bands 30, 50, and 21, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) GLT-Net (79.87%). (e) DSHFNet (71.09%). (f) M2FNet (72.57%). (g) FDNet (77.76%). (h) MICF-Net (75.35%). (i) S3F2Net (76.89%). (j) IDNet (74.08%). (k) CHNet (73.53%). (l) AdaSFMNet (80.25%).

allows the direct visual assessment of method performance relative to input data characteristics and ground reality.

2) Houston Dataset: The Houston dataset is known for its high scene complexity and diverse land cover types (with some areas occluded by clouds). From Table I, it can be seen that AdaSFMNet achieves the best performance on all objective metrics on this dataset. Compared with the second-best method GLT-Net, AdaSFMNet achieves improvements of 0.39%, 0.29%, and 0.42% in OA, AA, and Kappa, respectively. Although the improvements seem modest, this consistent lead demonstrates the effectiveness of the proposed method. Furthermore, AdaSFMNet exhibits strong competitiveness across

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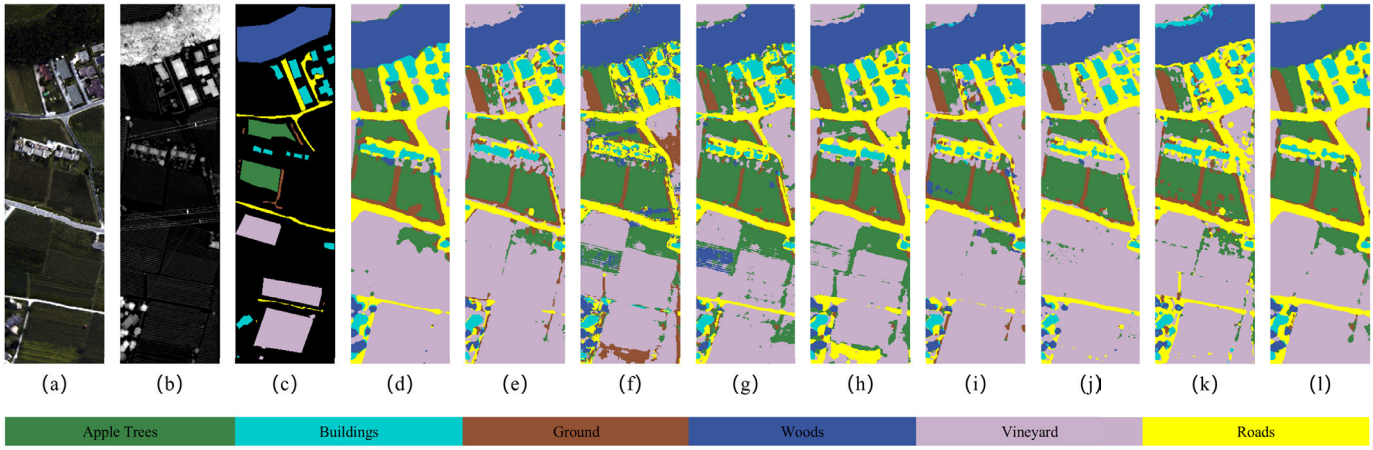


Fig. 8. Classification maps for all pixels in the Trento dataset using different methods. (a) False-color HSI (using bands 20, 15, and 5, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) GLT-Net (97.23%). (e) DSHFNet (98.33%). (f) M2FNet (95.98%). (g) FDNet (96.73%). (h) MICF-Net (97.54%). (i) S3F2Net (98.22%). (j) IDNet (98.19%). (k) CHNet (98.34%). (l) AdaSFMNet (98.75%).

many classes: among the 15 classes, it achieves the highest classification accuracy in one class (C10: Highway) and ranks in the top three in nine classes, with most classes exceeding 90% accuracy. Notably, the classes with good classification results typically exhibit clear spatial distribution and stable frequency-domain characteristics. For example, C5 (Soil), C14 (Tennis Court), and C10 (Highway) all achieve leading or near-leading accuracy. These classes often have well-defined structural boundaries or unique spectral responses, allowing the adaptive wavelet decomposition to precisely capture their frequency characteristics. Moreover, for natural classes with blurred boundaries, such as C1 (Healthy Grass) and C3 (Stressed Grass), although not reaching the highest accuracy, they remain at a high level, demonstrating the ability of AWFAM in collaboration with CNN to reduce confusion among similar classes. However, certain classes remain challenging for AdaSFMNet. For instance, C2 (Stressed Grass) achieves only 83.38%, trailing CHNet by over 6%. Similarly, C12 (Parking Lot 1) shows relatively lower accuracy compared with other classes. These classes typically lack distinct structural boundaries in LiDAR data and share similar spectral characteristics with adjacent categories. For example, Stressed Grass and Healthy Grass exhibit comparable spectral responses and elevation profiles, while Parking Lot 1 often resembles roads in terms of elevation. In such cases, the spatial–frequency joint modeling provides limited discriminative cues, as neither the low-frequency global contours nor the high-frequency edge details offer sufficient separation from visually similar classes. In the visual classification results (see Fig. 6), the classification maps generated by AdaSFMNet exhibit less intraclass noise and clearer structural features. In the magnified view, other compared methods (GLT-Net, DSHFNet, M2FNet, FDNet, S3F2Net, IDNet, and CHNet) fail to completely identify road features, showing multiple discontinuities. In contrast, AdaSFMNet, with its powerful spatial–frequency feature extraction capability and effective feature fusion algorithm, captures more discriminative representations, thereby achieving more continuous and complete road recognition. Additionally, some areas of this dataset are severely covered by clouds. Many compared methods rely more heavily on HSI information, leading to large classification errors in these areas. In contrast, AdaSFMNet exploits the immunity of LiDAR to cloud cover and, through the multimodal fusion method of SF-Mamba that removes redundant information and reduces HSI weights, effectively compensates for the information gap in optical images, thus achieving more accurate classification results in cloud-covered regions.

3) MUUFL Dataset: The MUUFL dataset contains 11 land cover classes with high spectral similarity among classes and disturbances, such as building shadows. From Table II, AdaSFMNet ranks first in OA (80.25%) and Kappa (74.62%), outperforming the second-best model by 0.38% and 0.15%, respectively; its AA (75.62%) ranks second, slightly lower than CHNet (75.86%) but still competitive overall. At the class level, AdaSFMNet achieves the highest accuracy on C6 (Water) with 100%, C8 (Building) with 91.05%, and C2 (Mostly Grass) with 71.87%. In particular, for C8 (Building), it improves by 1.71% over the second-best GLT-Net, and C6 (Water) reaches 100% accuracy. Moreover, the good performance of FDNet on C8 (Building), C6 (Water), and C2 (Mostly Grass) also fully demonstrates that frequency-domain features have excellent discriminative ability for such ground objects. However, on difficult classes such as C4 (Dirt and Sand), C9 (Sidewalk), and C10 (Yellow Curb), AdaSFMNet achieves notably lower accuracy than several compared methods. A closer examination reveals that these classes share two common challenges. First, with only ten training samples per class, the model lacks sufficient labeled data to learn reliable decision boundaries for these spectrally ambiguous categories. Second, these classes are often highly mixed with visually similar backgrounds—for instance, sidewalks and yellow curbs are typically adjacent to roads and share comparable spectral signatures and elevation profiles. As a result, when both the target class and its surrounding background exhibit similar low-frequency contours and high-frequency textures, the spatial–frequency features extracted by AWFAM provide limited discriminability, leading to frequent confusion at class boundaries. Nevertheless, the overall OA of AdaSFMNet still

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leads the second-best model, proving its robustness. In the visual comparison (see Fig. 7), S3F2Net misclassifies C7 (Building Shadow) as C1 (Trees) at the boundary between C1 (Trees) and C7 (Building Shadow). Although its classification accuracy for C1 (Trees) reaches the best, this comes at the cost of sacrificing accuracy for C7 (Building Shadow). In contrast, AdaSFMNet enhances the alignment between low-frequency structural information and high-frequency details through the cross-attention mechanism, generating clear and coherent class boundaries and significantly reducing misclassifications in shadowed areas.

4) Trento Dataset: The Trento dataset contains six land cover classes, with relatively simple scenes and high class separability. From Table III, AdaSFMNet achieves an OA of 98.75%, which is 0.41% higher than the second-best model CHNet; its AA reaches 97.30%, slightly lower than CHNet but still the best overall; and Kappa reaches 98.33%, 0.54% higher than CHNet. At the class level, AdaSFMNet achieves the highest accuracy on C4 (Woods) with 100%, C5 (Vineyard) with 100%, and C6 (Roads) with 99.56%. In particular, for C6 (Roads), it improves by 2.5% over the second-best MICFNet (97.06%), demonstrating the outstanding advantage of spatial–frequency joint modeling for linear structural objects. On C2 (Buildings), AdaSFMNet achieves slightly lower accuracy, likely due to the high intraclass spectral variability caused by diverse roof materials in the Trento dataset. Since AWFAM separates frequency components based on global statistics, it may not fully capture all subclass variations when frequency-domain features are widely distributed within a single category. Nevertheless, the impact on OA is small. In the visual results (see Fig. 8), AdaSFMNet generates smoother classification boundaries in the adjacent areas of C2 (Buildings) and C6 (Roads), whereas CHNet and M2FNet misclassify some buildings as roads. In summary, through the mutual validation of quantitative and visual analyses, the proposed AdaSFMNet demonstrates superior classification performance on three datasets with different complexities and sensor characteristics, fully proving the effectiveness and advancement of the AWFAM and the SF-Mamba modules at both shallow and deep fusion stages.

The above per-class analysis across the three datasets reveals both the strengths and limitations of AdaSFMNet. On the one hand, the collaborative design of AWFAM and SF-Mamba consistently contributes to improved classification accuracy. First, compared with methods that rely on shallow fusion, AdaSFMNet employs deeper cross-modal interaction through the cascaded shallow and deep SF-Mamba modules, allowing the complementary information of HSI and LiDAR to be more thoroughly exploited. Second, unlike methods that operate primarily in a single domain, such as GLT-Net and M2FNet which focus on the spatial domain, and FDNet which relies on global frequency filtering, AdaSFMNet benefits from a progressive spatial–frequency optimization architecture. AWFAM performs adaptive frequency decomposition tailored to each modality, while SF-Mamba jointly models local spatial dependencies and global frequency characteristics through its dual-branch design. This synergistic combination is particularly effective for classes with well-defined structural boundaries and distinct frequency responses, where both low-frequency contours and high-frequency edge details provide meaningful discriminative cues. On the other hand, several limitations remain. Performance tends to degrade for categories with extremely limited training samples or severe background mixing, where reliable spatial–frequency representations are difficult to learn. When a single category exhibits high intraclass spectral variability due to diverse material compositions, such as buildings with multiple roof types in Trento, the adaptive wavelet decomposition based on global statistics may not fully capture all subclass variations. Furthermore, although the Mamba backbone ensures linear complexity, the dual-branch spatial–frequency preprocessing in SF-Mamba introduces moderate additional computational cost compared with simpler fusion strategies.

In summary, through the mutual validation of quantitative and visual analyses, AdaSFMNet demonstrates superior classification performance on three datasets with different complexities and sensor characteristics, validating the effectiveness of its adaptive frequency decoupling and progressive spatial–frequency optimization design.

E. Ablation Studies

To evaluate the contribution of each core component, we conducted ablation studies on the three datasets, with results summarized in Table IV. Four variants are compared: 1) spatial-only, which relies solely on spatial feature extraction; 2) shallow fusion SF-Mamba, which applies SF-Mamba at the shallow fusion stage to perform joint spatial–frequency optimization on the representations encoded by AWFAM; 3) deep fusion SF-Mamba, which applies SF-Mamba at the deep fusion stage to refine the cross-domain fused spatial–frequency features, enhancing noise robustness; and 4) proposed method, which incorporates all proposed modules. All variants are evaluated under identical parameter settings.

The OA, AA, and Kappa results of the four variants are reported in Table IV. Across all three datasets, the methods incorporating frequency-domain information consistently outperform the spatial-only baseline, with particularly significant improvements on Houston2013 and MUUFL. This confirms that the frequency-domain information extracted by AWFAM provides critical complementary cues beyond pure spatial features, enabling the model to more fully exploit the complementary information of HSI and LiDAR data and thereby enhancing its discriminative ability. These findings underscore the robust and versatile role of AWFAM in capturing rich frequency information for AdaSFMNet. A notable performance gap in OA between the Shallow fusion SF-Mamba variant, and the full model is observed on all datasets, which we attribute to two factors. First, the 2D-SSD module is inherently well-suited for processing 2-D image features and effectively models long-range dependencies in spatial–frequency joint representations, bringing substantial benefits to the HSI–LiDAR joint classification task. Second, the spatial perception branch $S(\cdot)$ strengthens local contextual dependencies, while the frequency filtering branch $T(\cdot)$ adaptively suppresses noise and redundant frequencies, facilitating the complementary integration of global frequency information. Although the deep fusion

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TABLE IV
ABLATION STUDY OF KEY COMPONENTS ON THREE DATASETS. BEST RESULTS ARE IN BOLD

Method	Houston2013			MUUFL			Trento		
	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
Spatial-only	89.75	90.59	88.92	79.77	**77.05**	74.15	98.67	96.94	98.22
Shallow fusion SF-Mamba	91.37	91.42	90.67	78.68	74.29	72.72	97.94	96.07	97.25
Deep fusion SF-Mamba	91.68	92.34	91.01	79.02	71.32	72.79	98.67	97.27	98.23
Proposed method	**91.85**	**92.22**	**91.19**	**80.25**	75.62	**74.62**	**98.75**	**97.30**	**98.33**

SF-Mamba variant achieves competitive results, it still falls behind the full model. This indicates that while the deep state-space module effectively suppresses noise and balances local details with global structure, the absence of shallow cross-domain optimization on the AWFAM-encoded spatial–frequency joint representations allows noise and redundant frequencies to persist into later fusion stages, limiting the model's ability to handle complex cross-domain redundancy.

Overall, the full AdaSFMNet achieves the best performance across all datasets and metrics, demonstrating the synergistic contribution of AWFAM, shallow SF-Mamba, and deep SF-Mamba in jointly modeling spatial and frequency information for HSI–LiDAR classification.

V. CONCLUSION

This article proposes a novel deep learning model, AdaSFMNet, for HSI and LiDAR data classification. The framework adaptively matches the differentiated frequency band characteristics of HSI and LiDAR while progressively optimizing and fusing spatial–frequency mixed features. In the AWFAM module, wavelet kernels are parameterized as learnable convolution operators to extract frequency-domain features, and the semantics of low-frequency and high-frequency features are aligned via a cross-attention mechanism. The SF-Mamba enhances local dependencies through depthwise convolution in the spatial branch and eliminates redundant information via adaptive Fourier filtering in the frequency branch, thereby strengthening global representation capability. This equips Mamba with complete spatial–frequency perception and enables local–global modeling. Extensive experiments are conducted on three datasets, including the challenging Houston2013 dataset. Quantitative and visual analyses demonstrate that the proposed AdaSFMNet outperforms other related networks. Moreover, the fusion strategy of multimodal data is crucial for the classification of HSI and LiDAR data. Therefore, a main direction for future research is to develop more effective adaptive multimodal feature fusion mechanisms.

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